

OFBD: Object-Focused Background Debiasing for Long-tailed Learning

Overview: This supplementary material provides additional mathematical derivations (Appendix B), extensive visualization results (Appendix C), additional experimental results and ablations (Appendix D), and comprehensive implementation details (Appendix A) to support the main paper.

A More Experiment Protocols

A.1 Implementation Details

OFBD is implemented as a plug-and-play module and can be integrated into various long-tailed recognition frameworks without altering their original backbones or classifier designs.

For CIFAR-10-LT and CIFAR-100-LT, we use ResNet-32 as the backbone. Models are trained for 200 epochs with a batch size of 256 using SGD with momentum 0.9 and weight decay 5×10^{-4} . The learning rate is linearly warmed up to 0.15 during the first 20 epochs and decayed by 0.1 at epochs 160 and 180. When OFBD is plugged into contrastive-learning baselines, *e.g.*, BCL [110] and SBCL [22], we keep their original contrastive objectives unchanged. The projection dimension is 128, the contrastive temperature is 0.1, and the classification and contrastive losses are weighted by 2 and 0.6, following the original baseline settings [22, 110]. For non-contrastive baselines, the projection head and contrastive loss are removed.

For FG-CutMix, we apply the augmentation with probability 0.5. Given an input image, we randomly generate a region proposal set $\mathcal{P}(x) = \{b_1, \dots, b_N\}$ with different locations and scales. Instead of ranking proposals by local energy, the RL-based region selector π_θ predicts a selection probability over $\mathcal{P}(x)$, and K proposals are sampled without replacement to form the foreground candidate pool $\mathcal{R}_\theta(x) = \{r_1, \dots, r_K\}$. For each selected candidate, we construct a mixed sample by preserving the selected target-related region and replacing its complementary region with content from another image. The mixed label is corrected in line with prior CutMix-based augmentation methods.

For selector optimization, the classifier prediction on each mixed sample provides the reward signal. Specifically, the reward is set to 1 if the mixed sample has a higher predicted probability for the target foreground-source class than for the background-source class, and 0 otherwise. Rewards are normalized within each candidate pool to compute relative advantages. The selector is optimized by policy gradient with a clipped objective and KL regularization to a frozen reference selector π_{ref} , which is initialized from the previous epoch selector to stabilize policy updates. Unless otherwise specified, we set the RL loss weight $\gamma = 0.8$, the auxiliary selector supervision weight $\alpha = 1.0$, the KL coefficient $\beta = 0.02$, and the clipping range $\epsilon_c = 0.2$.

For BFR, we use the final convolutional feature map as input. Neuron-wise importance scores are computed from channel-wise feature statistics. Effective channels are selected by comparing each neuron-wise score with the spatial mean score, and the resulting foreground contribution is used to derive background-aware scores. The background-biased component is then estimated by background-aware weighted aggregation, and the rectified representation is computed as $f' = f - \lambda f_{\text{bg}}^{\text{BFR}}$, where $\lambda = 0.5$ in all experiments. The rectified representation is fed into the classifier and, when applicable, the projection head. BFR introduces no learnable parameters and requires no external annotations, pretrained localization models, or additional supervision.

For ImageNet-LT and iNaturalist 2018, we use ResNet-50 [20] and ViT-B/16 [14] as backbones. Models are trained with SGD, momentum 0.9, and batch size 256. For ImageNet-LT, we train for 90 epochs with a learning rate of 0.1 and weight decay 5×10^{-4} . For iNaturalist 2018, we train for 100 epochs with a learning rate of 0.2 and a weight decay of 1×10^{-4} . Unless otherwise specified, the same OFBD plug-in hyperparameters are used across datasets.

A.2 Dataset and Evaluation Protocol

We evaluate OFBD on CIFAR-10-LT, CIFAR-100-LT, ImageNet-LT, and iNaturalist 2018. For CIFAR-10-LT and CIFAR-100-LT, we use imbalance ratios $r \in \{100, 50, 10\}$. The original balanced test sets are used for evaluation. For ImageNet-LT and iNaturalist 2018, we follow the official

train/validation splits [101]. We report Top-1 accuracy for all datasets. For class-frequency analysis, we report Many-, Medium-, and Few-shot accuracy, where classes with more than 100 samples are Many-shot, classes with 20–100 samples are Medium-shot, and classes with fewer than 20 samples are Few-shot. For plug-in comparisons, OFBD keeps the official baseline’s backbone, objective, classifier, sampler, and training schedule.

B Theoretical Derivations and Analysis

B.1 Interpretation of Distribution-level Background Bias (Eq. 1)

In Sec. 3, we hypothesized that tail classes suffer from a larger background distribution shift in Eq. 1 and further corroborate this hypothesis through empirical observations. Here, we provide an interpretation to mathematically substantiate this claim based on sample complexity, deeply aligned with findings on spurious causal correlations in imbalanced data [77]. Let N_y denote the number of training samples for class y . In a long-tailed recognition, we have $N_{head} \gg N_{tail}$.

1. Deviation of Empirical Distributions.

Let q_y denote the ideal background distribution for class y constructed from a fully balanced, infinite-sample dataset. Under the long-tailed setting, the model observes an empirical background distribution, denoted as p_y , which is estimated from only N_y samples.

According to statistical learning theory [4, 109, 42], the deviation between an empirical distribution and the true underlying distribution is bounded by the sample size. Specifically, the expected L_1 distance between p_y and q_y converges at a rate of $\mathcal{O}(1/\sqrt{N_y})$:

$$\mathbb{E}[\|p_y - q_y\|_1] \leq \mathcal{O}\left(\frac{1}{\sqrt{N_y}}\right). \quad (11)$$

Since $N_{head} \gg N_{tail}$, the expected distribution deviation for tail classes is significantly larger than that for head classes [4]. This provides the theoretical basis for the inequality:

$$\|p_y^{tail} - q_y\|_1 \gg \|p_y^{head} - q_y\|_1. \quad (12)$$

2. Bound on Background Feature Bias.

Next, we relate this distribution-level deviation to the feature-level background bias $\Delta D_{bg}(y)$. By definition, the expected background feature means are

$$\mu_{bg}(y) = \mathbb{E}_{b \sim p_y}[f_{bg}], \quad \bar{\mu}_{bg}(y) = \mathbb{E}_{b \sim q_y}[f_{bg}]. \quad (13)$$

The background bias $\Delta D_{bg}(y)$ can be expanded as

$$\begin{aligned} \|\Delta D_{bg}(y)\|_1 &= \|\mu_{bg}(y) - \bar{\mu}_{bg}(y)\|_1 \\ &= \left\| \int f_{bg} p_y(b) db - \int f_{bg} q_y(b) db \right\|_1 \\ &= \left\| \int f_{bg} (p_y(b) - q_y(b)) db \right\|_1 \end{aligned} \quad (14)$$

Assuming the feature representations f_{bg} extracted by a standard deep neural network are inherently bounded (e.g., strictly constrained by non-linear activation functions or normalization layers [27, 3]), there exists a constant $M > 0$ such that $\sup_b \|f_{bg}\|_1 \leq M$. Using Hölder’s inequality, we obtain an upper bound on the feature bias:

$$\|\Delta D_{bg}(y)\|_1 \leq \int \|f_{bg}\|_1 \cdot |p_y(b) - q_y(b)| db \leq M \int |p_y(b) - q_y(b)| db = M \|p_y - q_y\|_1. \quad (15)$$

This bound indicates that larger background distribution deviation can increase the possible magnitude of background feature shift, which supports the hypothesis in Eq. 1. In particular, since tail classes have fewer samples, their empirical background distributions can deviate more from the balanced reference, potentially leading to larger background-biased feature shifts compared with head classes. Hence, this motivates a targeted debiasing mechanism for tail classes, as standard empirical risk minimization (ERM) may overfit spurious background correlations for minority categories [67].

B.2 Formal Analysis of Optimization-level Background Bias (Eq. 3)

We provide a concise justification for Eq. 3 from spectral bias and implicit low-rank regularization [2, 7]. Following Sec. 3.1, we decompose the feature as $f = f_{\text{fg}} + f_{\text{bg}}$, and define:

$$R_t(y) = \frac{1}{1 + \gamma_t(y)}, \quad \gamma_t(y) = \frac{G_{\text{fg},t}(y)}{G_{\text{bg},t}(y)}. \quad (16)$$

Spectral bias and low-rank learning dynamics. Under gradient descent, deep networks exhibit spectral bias, favoring directions with larger singular values [2]. Let $a_k(t)$ denote the alignment of the model with the k -th spectral component. Its evolution approximately follows:

$$\frac{d}{dt} a_k(t) \propto \sigma_k^2 a_k(t), \quad (17)$$

which leads to exponentially faster growth along dominant directions. Empirically, deep representations exhibit low-rank structures during training [7], meaning that most energy concentrates in a few principal components. Background features, due to spatial redundancy and shared patterns across samples, naturally form a low-rank structure with dominant singular values. In contrast, foreground features are more diverse and distributed across many weaker components. As a result, gradient descent preferentially fits background-related components early in training, establishing an inherent optimization bias toward low-rank structures.

Effect of sample size on effective rank. For class y , the effective rank of foreground features depends on the sample size N_y . When N_y is large (head classes), diverse foreground patterns are observed, allowing multiple spectral components of f_{fg} to be learned. In contrast, for tail classes with limited samples, the observed foreground variations are restricted, reducing the usable rank of f_{fg} . This further amplifies the low-rank bias: the background components remain stable due to cross-class sharing, while the high-rank foreground structure becomes increasingly difficult to capture. Consequently, optimization for tail classes gradually shifts toward low-rank (background) directions.

Evolution of gradient ratio. Combining spectral bias and effective-rank reduction, the relative strength between foreground and background gradients evolves multiplicatively over time. Following standard analyses of gradient flow in linearized regimes [2], the ratio $\gamma_t(y)$ can be approximated as:

$$\gamma_t(y) \approx \gamma_0(y) \exp((\sigma_{\text{fg}}^2 - \sigma_{\text{bg}}^2)t), \quad (18)$$

where the exponent reflects the difference in effective learning rates. For tail classes, the reduced effective rank suppresses foreground components, leading to $\sigma_{\text{fg}}^2 < \sigma_{\text{bg}}^2$, and thus $\gamma_t(y) \rightarrow 0$. Substituting into $R_t(y)$ gives $R_t(y) \rightarrow 1$, implying $\Delta R_t(y) > 0$. For head classes, sufficient samples preserve the foreground rank and prevent $\gamma_t(y)$ from collapsing. Meanwhile, background components are fitted early and their gradients diminish over time, leading to a non-increasing $R_t(y)$ (i.e., $\Delta R_t(y) \simeq 0$). Therefore:

$$\Delta R_t^{\text{tail}} > \Delta R_t^{\text{head}}. \quad (19)$$

B.3 Detailed Derivation of $S_{d,u}$ in Eq. 6

In long-tailed visual recognition, models tend to overfit to background feature, particularly for tail classes. Addressing this optimization-level bias typically requires extra modules. However, introducing learnable parameters often exacerbates class imbalance due to biased gradient updates. To rectify this without additional parameters, we propose estimating the inherent foreground distinctiveness of each neuron based on its feature statistics. Inspired by the energy-based local spatial formulation in SimAM [96], we hypothesize that target-related foreground features exhibit distinctive activation patterns compared to the widely distributed, background feature. To quantify this foreground distinctiveness, we measure the linear separability between the target neuron $F_{d,u}$ and the remaining background neurons $F_{d,v}$ ($v \neq u$). Assigning a target label of 1 to $F_{d,u}$ and -1 to all other neurons, the energy function to optimize the linear transform weights (w, b) is formulated as:

$$e(w, b) = \frac{1}{M-1} \sum_{v \neq u} (-1 - (wF_{d,v} + b))^2 + (1 - (wF_{d,u} + b))^2 + \eta w^2, \quad (20)$$

where ηw^2 serves as an L_2 regularizer. By taking the partial derivatives of the objective with respect to w and b and setting them to zero, we obtain a fast closed-form solution:

$$w^* = -\frac{2(F_{d,u} - \mu_t)}{(F_{d,u} - \mu_t)^2 + 2\sigma_t^2 + 2\eta}, \quad b^* = -\frac{1}{2}(F_{d,u} + \mu_t)w^*, \quad (21)$$

where μ_t and σ_t^2 are the mean and variance computed over all neurons in the channel except the target $F_{d,u}$. To circumvent the computational overhead of iterative spatial calculations, we approximate them using the global channel mean μ_d and variance σ_d^2 calculated over all M neurons. Substituting w^* and b^* back yields the minimal energy e^* :

$$e^* = \frac{4(\sigma_d^2 + \eta)}{(F_{d,u} - \mu_d)^2 + 2\sigma_d^2 + 2\eta}. \quad (22)$$

A lower minimal energy e^* mathematically indicates that the target neuron $F_{d,u}$ strongly deviates from the background-dominated channel context. While prior work treats this merely as general visual saliency [96, 92, 6], we repurpose it as a highly reliable, unsupervised proxy for foreground information in imbalanced learning. Thus, the inherent foreground contribution of the neuron can be represented by the inverse of the minimal energy ($1/e^*$):

$$\frac{1}{e^*} = \frac{(F_{d,u} - \mu_d)^2 + 2\sigma_d^2 + 2\eta}{4(\sigma_d^2 + \eta)} = \frac{(F_{d,u} - \mu_d)^2}{4(\sigma_d^2 + \eta)} + 0.5. \quad (23)$$

By omitting the constant scaling offset (0.5) and replacing the regularizer η with ϵ for numerical stability, we map this dynamic distinctiveness through a sigmoid function. This directly yields the bounded score $S_{d,u}$ utilized in our main framework (Eq. 6):

$$S_{d,u} = \text{sigmoid}\left(\frac{(F_{d,u} - \mu_d)^2}{4(\sigma_d^2 + \epsilon)}\right). \quad (24)$$

Additionally, we emphasize a fundamental distinction from conventional attention modules [99, 30] regarding **gradient bias and parameterization**. Standard attention modules rely on *learnable* parameters to generate dynamic scores. Under severe long-tailed imbalance, the gradient updates for these parameters are overwhelmingly dominated by head classes. Consequently, to minimize the overall empirical risk, these modules inevitably learn to attend to and universally amplify spurious head-class backgrounds, inadvertently acting as “bias amplifiers”. In contrast, our approach derives $S_{d,u}$ purely from the *intrinsic spatial statistics* of the feature map, rendering it strictly parameter-free and immune to class-imbalanced gradient domination. By aggregating these statistical scores into a deterministic background penalty map B_u (Eq. 7), our BFR module shifts from generic feature reweighting to explicit spatial decoupling. It mathematically penalizes spurious background-foreground co-occurrences (Eq. 9), preventing tail-class representations from collapsing into dominant background features.

B.4 Why OFBD Mitigates Background Bias

We further justify that the two components of OFBD directly reduce the two bias terms derived in Sec. B.1 and Sec. B.2. Specifically, FG-CutMix mitigates the distribution-level bias in Eq. 1, while BFR suppresses the optimization-level bias in Eq. 2.

FG-CutMix reduces distribution-level background Bias. From Sec. B.1, the background feature bias is upper-bounded by the background distribution deviation. Thus, it is sufficient to show that FG-CutMix reduces $\|p_y - q_y\|_1$. Let $\tilde{p}_y$ denote the effective background distribution induced by FG-CutMix. Since FG-CutMix preserves the target foreground and replaces the complementary background, $\tilde{p}_y$ can be written as:

$$\tilde{p}_y = (1 - \eta_y)p_y + \eta_y p_{\text{mix}}, \quad \eta_y \in [0, 1], \quad (25)$$

where p_{mix} denotes the background distribution introduced by mixed samples. Since p_{mix} aggregates backgrounds from more samples, it provides a closer estimate of the ideal background distribution:

$$\|p_{\text{mix}} - q_y\|_1 \leq \|p_y - q_y\|_1. \quad (26)$$

By the convexity of the L_1 norm:

$$\|\tilde{p}_y - q_y\|_1 \leq (1 - \eta_y)\|p_y - q_y\|_1 + \eta_y\|p_{\text{mix}} - q_y\|_1 \leq \|p_y - q_y\|_1. \quad (27)$$

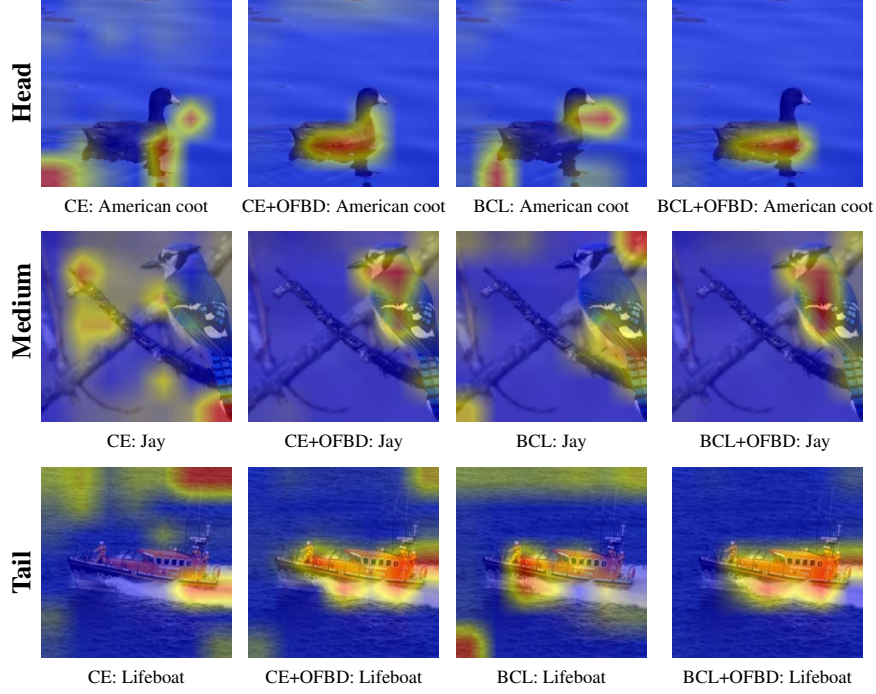


Figure 4: Extended pretrained-CAM attention maps of representative **Head, Medium, and Tail** classes under CE, CE+OFBD, BCL, and BCL+OFBD. OFBD reduces background co-activation and makes attention more object-focused across different baselines and class-frequency groups.

Therefore, using the bound established in Sec. B.1, we obtain:

$$\|\Delta \tilde{D}_{bg}(y)\|_1 \leq \|\Delta D_{bg}(y)\|_1. \quad (28)$$

This shows that FG-CutMix reduces the upper bound of distribution-level background bias. Moreover, the RL reward in Eq. 4 favors regions whose mixed samples remain more predictive of the target label than the background-source label, preventing the mixed distribution from being dominated by spurious background semantics.

BFR reduces optimization-level background bias. From Sec. B.2, the optimization-level bias is characterized by the increase of the background gradient ratio. According to Eq. 9, BFR rectifies the feature by assigning each spatial location a background-aware weight $(1 - \lambda B_u)$. Since a larger B_u indicates stronger background bias, background-dominant locations receive smaller weights than foreground-dominant locations:

$$B_u^{bg} > B_v^{fg} \Rightarrow 1 - \lambda B_u^{bg} < 1 - \lambda B_v^{fg}. \quad (29)$$

Therefore, the background-related feature component, and hence its gradient contribution, is more strongly suppressed than the foreground component:

$$\|\tilde{g}_{bg,t}(y)\|_1 \leq \|g_{bg,t}(y)\|_1, \quad \frac{\|\tilde{g}_{bg,t}(y)\|_1}{\|g_{bg,t}(y)\|_1} < \frac{\|\tilde{g}_{fg,t}(y)\|_1}{\|g_{fg,t}(y)\|_1}. \quad (30)$$

Substituting the rectified gradients into the background gradient ratio defined in Eq. 2 yields

$$\tilde{R}_t(y) = \frac{\|\tilde{g}_{bg,t}(y)\|_1}{\|\tilde{g}_{fg,t}(y)\|_1 + \|\tilde{g}_{bg,t}(y)\|_1} \leq \frac{\|g_{bg,t}(y)\|_1}{\|g_{fg,t}(y)\|_1 + \|g_{bg,t}(y)\|_1} = R_t(y). \quad (31)$$

Thus,

$$\tilde{R}_t(y) \leq R_t(y), \quad \Delta \tilde{R}_t(y) \leq \Delta R_t(y). \quad (32)$$

This shows that BFR suppresses the optimization-level background shift by reducing the relative contribution of background gradients.

Table 9: Ablation on hyper-parameters β and α .

β	Acc. (%)	α	Acc. (%)
0.005	53.7	0.1	53.6
0.01	53.9	0.5	53.8
0.015	54.0	0.75	54.1
0.02	54.2	1.0	54.2
0.025	54.1	1.25	53.6
0.03	53.8	1.5	53.7

Table 10: Ablation on hyper-parameters λ and γ .

λ	Acc. (%)	γ	Acc. (%)
0.2	53.6	0.2	53.3
0.3	53.9	0.4	53.7
0.4	53.7	0.6	54.1
0.5	54.2	0.8	54.2
0.6	54.1	1.0	53.8
0.7	53.9	1.2	53.4

Table 11: Background distribution shift on CIFAR-100-LT ($r = 100$). Lower is better.

Method	Many↓	Med.↓	Few↓	All↓
CE	0.79	0.84	0.86	0.83
+ FG-CutMix	0.78	0.83	0.80	0.82
+ BFR	0.79	0.84	0.81	0.80
+ OFBD	0.80	0.79	0.78	0.79

Table 12: Background gradient ratio shift on CIFAR-100-LT ($r = 100$). Lower is better.

Method	Many↓	Med.↓	Few↓	All↓
CE	0.4	0.8	3.6	1.5
+ FG-CutMix	-3.4	-4.7	-6.2	-4.7
+ BFR	-4.6	-5.2	-6.5	-5.0
+ OFBD	-5.6	-6.1	-7.0	-6.2

Joint effect. Combining the above results, OFBD reduces both bias terms:

$$\|\Delta\tilde{D}_{bg}(y)\|_1 \leq \|\Delta D_{bg}(y)\|_1, \quad \Delta\tilde{R}_t(y) \leq \Delta R_t(y). \quad (33)$$

Hence, FG-CutMix mitigates the enlarged background distribution deviation of tail classes, while BFR suppresses their amplified background gradient. This provides a formal explanation for why OFBD alleviates both distribution-level and optimization-level background bias in long-tailed recognition.

C Extensive Visualizations

C.1 Extended Background-bias Visualization

To further support the analysis in Sec. 5.3, we provide extended attention-map visualizations for representative head, medium, and tail classes. As shown in Fig. 4, we compare CE, CE+OFBD, BCL, and BCL+OFBD using pretrained-CAM attention maps. The selected samples cover different class-frequency groups, including head, medium, and tail classes. For both CE and BCL baselines, the attention maps often co-activate target objects and surrounding backgrounds, indicating background-biased representation. After applying OFBD, the activated regions become more concentrated on target-related foregrounds, while irrelevant background responses are reduced. This trend is observed across different baselines and class-frequency groups, suggesting that OFBD consistently improves object-focused representation rather than only benefiting a specific model or class group. These visualizations further validate that OFBD alleviates background reliance.

D Additional Experimental Results

D.1 Hyper-parameters Ablation

We conduct hyper-parameter ablations on CIFAR-100-LT with imbalance ratio 100 to analyze the sensitivity of OFBD. As shown in Tables 9 and 10, OFBD remains stable across a wide range of hyper-parameter values. For the RL-based foreground selector, β controls the KL regularization strength and α controls the auxiliary selector supervision. The best performance is achieved at $\beta = 0.02$ and $\alpha = 1.0$, both reaching 54.2% accuracy. Smaller values provide insufficient regularization or supervision, while overly large values slightly degrade performance.

For the overall OFBD framework, λ controls the strength of background feature rectification and γ controls the weight of policy learning. The best accuracy is obtained at $\lambda = 0.5$ and $\gamma = 0.8$, both reaching 54.4%. When λ is too small, background-biased features are insufficiently suppressed; when

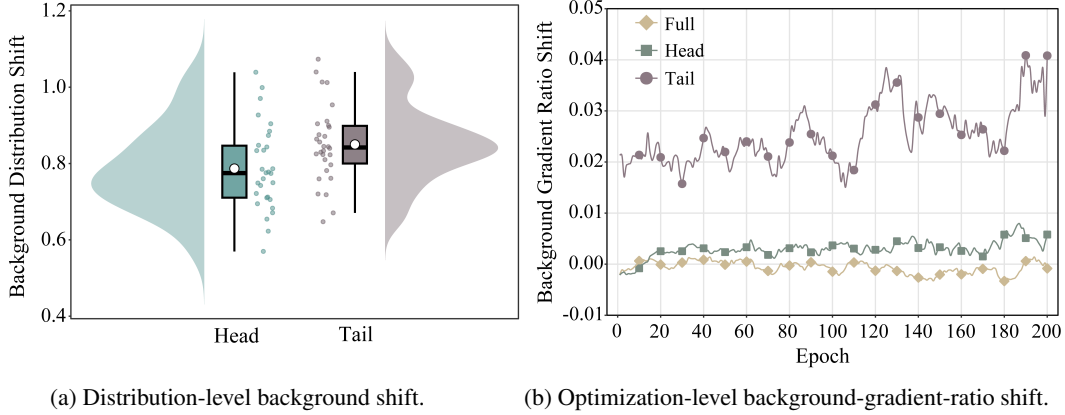


Figure 5: Empirical verification of background bias on CIFAR-100-LT with imbalance ratio 100. Tail classes exhibit both larger background distribution shift and stronger background-gradient-ratio shift than head classes, validating the distribution-level and optimization-level bias analyzed in Sec. 3.

too large, useful foreground information may also be weakened. Similarly, a moderate γ provides the best trade-off between selector learning and classification optimization. These results show that OFBD is not sensitive to exact hyper-parameter choices and maintains competitive performance under nearby settings.

D.2 Ablation on Candidate Pool Size

We further study the effect of the candidate pool size K in FG-CutMix, where K denotes the number of selected foreground candidates used for reward comparison and relative-advantage estimation. Small K values limit candidate diversity, reducing the selector’s ability to distinguish regions with different target-preservation quality. Increasing K improves the chance of selecting foregrounds that better preserve target semantics after background replacement. However, excessively large K increases computation cost and may introduce redundant or noisy candidates.

Fig. 6a visualizes the accuracy and per-epoch computation cost for different K on CIFAR-100-LT with imbalance ratio 100. Accuracy improves as K increases from 2 to 6, then slightly decreases for larger K , while per-epoch cost rises monotonically. This indicates that a moderate candidate pool provides sufficient diversity for effective foreground selection. We set $K = 6$ as the default, achieving a favorable trade-off between accuracy and efficiency.

D.3 Empirical Verification of Background Bias

Experiment Setting. We train ResNet-32 [20] with cross-entropy on CIFAR-100-LT [4] and CIFAR-100 under identical settings. The CIFAR-100 model serves as the balanced reference for measuring distribution-level background shift. Background regions are obtained with a pretrained CAM [106, 31].

Results. As shown in Fig. 5, tail classes suffer from stronger background bias than head classes. Specifically, Fig. 5a shows that tail classes have a larger background distribution shift, indicating that their learned background statistics deviate more from the balanced reference distribution. Meanwhile, Fig. 5b shows that the background-gradient-ratio shift of tail classes increases faster and fluctuates more strongly during training, suggesting that tail-class optimization is more easily driven by background-related gradients. These results support our analysis (Sec. 3.4) that tail degradation is not only caused by sample scarcity, but also by background-biased distribution and optimization.

D.4 Effect on Background Distribution Shift

Based on the observed distribution-level bias, we further evaluate whether FG-CutMix can reduce the background distribution shift. We measure the deviation between the background feature distribution learned under long-tailed training and that learned from balanced training. Specifically, we train a

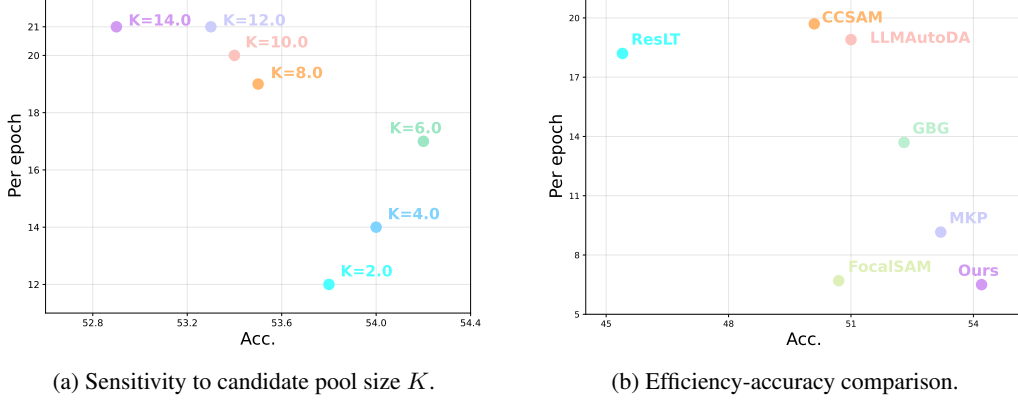


Figure 6: Efficiency analysis of OFBD on CIFAR-100-LT with imbalance ratio 100. (a) Accuracy and per-epoch (s) computation cost as a function of the candidate pool size K . Accuracy first improves and then slightly decreases for large K , while per-epoch cost rises monotonically. $K = 6$ is selected as the default for a good efficiency-accuracy trade-off. (b) Efficiency-accuracy comparison with representative long-tailed methods, where per-epoch computation cost is measured in seconds. “Ours” denotes BCL+OFBD.

reference model on CIFAR-100 and use it to estimate the class-wise reference background feature mean $\bar{\mu}_{bg}(y)$. For each long-tailed model, we compute the background feature mean $\mu_{bg}(y)$ on the same balanced test set, where background regions are obtained by pretrained CAM masks. The background distribution shift is measured as:

$$\Delta D_{bg}(y) = \|\mu_{bg}(y) - \bar{\mu}_{bg}(y)\|_1. \quad (34)$$

We report the averaged $\Delta D_{bg}(y)$ over Many-, Medium-, and Few-shot classes. Lower values indicate a smaller distribution-level background shift.

As shown in Table 11, standard long-tailed training exhibits a clear background distribution shift, especially on Few-shot classes. Standard CutMix only provides limited reduction, since random region replacement may still preserve or introduce spurious background correlations. In contrast, FG-CutMix consistently reduces $\Delta D_{bg}(y)$, indicating that preserving target-related foregrounds while diversifying backgrounds makes the learned background distribution closer to the balanced reference distribution.

D.5 Effect on Background Gradient Ratio

Based on the observed optimization-level bias, we further examine whether BFR can suppress the increase of the background gradient ratio. We measure the increase of the background gradient ratio defined in Eq. 2. Foreground and background gradient components are computed using the same foreground-background masks as in Sec. D.3. We report the averaged $\Delta R_t(y)$ over Many-, Medium-, and Few-shot classes. Lower values indicate that the optimization process is less dominated by background-related gradients.

As shown in Table 12, both FG-CutMix and BFR substantially reduce the background gradient ratio compared with standard CE, with the effect most pronounced for Few-shot classes. FG-CutMix decreases the gradient shift at the input level by diversifying backgrounds, while BFR further down-weights background-biased spatial features during feature-level rectification. The full OFBD, combining FG-CutMix and BFR, achieves the largest reduction across all class groups, demonstrating the complementarity between input-level background diversification and feature-level rectification in mitigating background-driven optimization.

D.6 Performance Analysis under Long-tailed and Balanced Settings

We further compare the head-, medium-, and tail-class performance to examine where OFBD brings the largest improvement. As shown in Fig. 8, adding OFBD to BCL substantially improves Few-shot accuracy from 32.9% to 40.2%, yielding a +7.3% gain. Meanwhile, it slightly improves

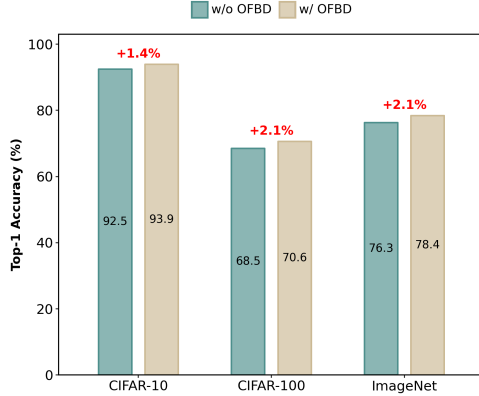


Figure 7: Performance comparison with and without OFBD on balanced datasets ($r = 1$). OFBD consistently improves Top-1 accuracy on CIFAR-10, CIFAR-100, and ImageNet.

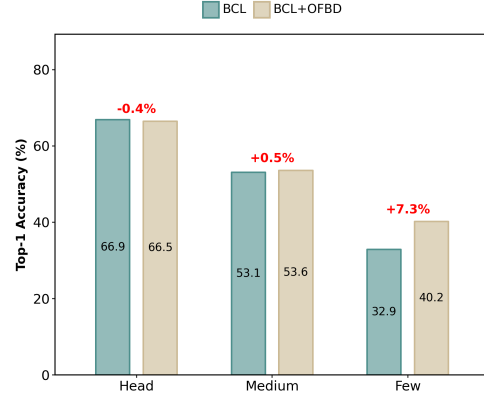


Figure 8: Head-, medium-, and tail-class performance comparison on CIFAR-100-LT with imbalance ratio 100. OFBD mainly improves tail-class accuracy while largely preserving head-class performance.

Medium-shot accuracy from 53.1% to 53.6%, with only a marginal change on Head classes from 66.9% to 66.5%. These results indicate that OFBD mainly benefits data-scarce tail classes while largely preserving head-class performance, demonstrating its effectiveness in alleviating tail-specific background bias.

We also examine whether OFBD is effective under the balanced setting, where the imbalance ratio is 1. As shown in Fig. 7, OFBD consistently improves the baseline without OFBD on balanced CIFAR-10, CIFAR-100, and ImageNet, with gains of +1.4%, +2.1%, and +2.1%, respectively. This demonstrates that OFBD does not rely on long-tailed class distributions and can still provide object-focused regularization benefits when the class distribution is uniform.

D.7 Further Ablation of FG-CutMix

Table 4 shows that FG-CutMix with RL-based foreground selection outperforms standard CutMix with random selection. To further analyze this improvement, we compare their optimization-level background bias by measuring the background-gradient-ratio shift on CIFAR-100-LT with imbalance ratio 100. As shown in Fig. 9a, both CutMix and FG-CutMix keep the head-class shift relatively small, suggesting that head-class optimization is less sensitive to region selection.

For tail classes, however, standard CutMix exhibits a stronger background-gradient-ratio shift, indicating that random replacement may preserve background cues or discard target-related foreground evidence. In contrast, FG-CutMix consistently produces a lower tail-class shift during training. This confirms that RL-based foreground region selection not only improves accuracy, as shown in Table 4, but also reduces background-driven optimization for tail classes.

We further examine the training stability of the RL-based foreground selector. As shown in Fig. 9b, the selector loss exhibits stochastic fluctuations due to random proposal sampling and policy-gradient optimization, but remains bounded and gradually stabilizes throughout training. This suggests that the selector does not suffer from unstable policy updates. Together with the accuracy improvement in Table 4 and the reduced tail-class background-gradient shift in Fig. 9a, these results indicate that the RL-based selector can be optimized stably while learning foreground regions that better mitigate background-driven optimization.

D.8 Efficiency Analysis

We evaluate the efficiency-accuracy trade-off of BCL+OFBD on CIFAR-100-LT with imbalance ratio 100. As shown in Fig. 6b, BCL+OFBD achieves the highest accuracy while maintaining a low per-epoch training cost. In particular, compared with MKP, BCL+OFBD reduces the computation cost by 2.66 seconds per epoch and improves Top-1 accuracy by 1.0%. These results show that OFBD

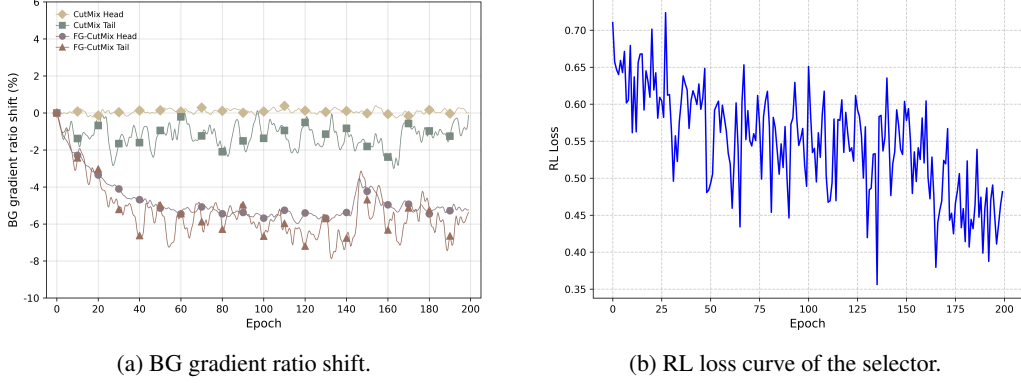


Figure 9: Further analysis of FG-CutMix on CIFAR-100-LT with imbalance ratio 100. (a) FG-CutMix reduces tail-class background-gradient-ratio shift compared with standard CutMix, indicating weaker background-driven optimization. (b) The RL loss of the foreground selector remains bounded and gradually stabilizes during training, indicating stable optimization of the RL-based selector.

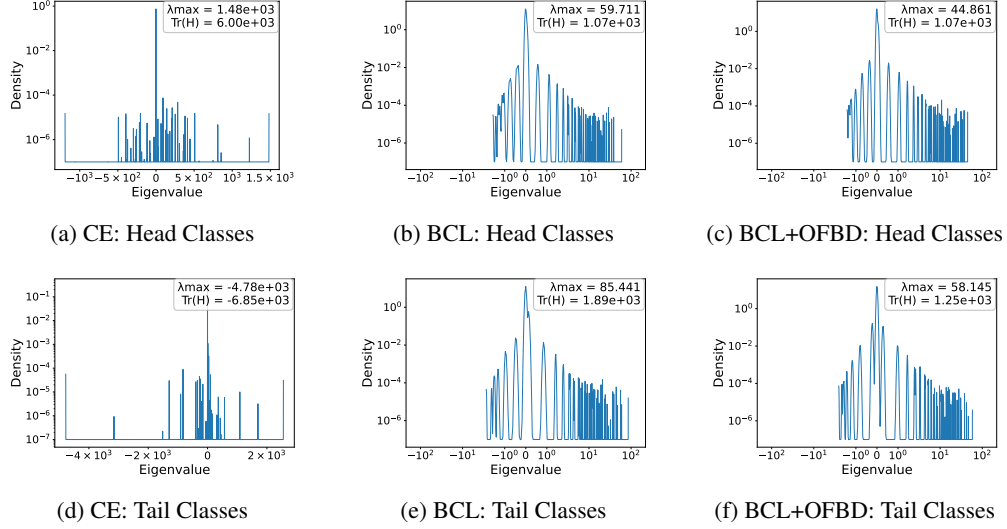


Figure 10: Eigen Spectral Density of Hessian for head and tail classes of ResNet models trained with CE, BCL, and BCL+OFBD on CIFAR-100 LT respectively. A smaller $\lambda_{\max}$ and $Tr(H)$ generally indicate a flatter loss landscape.

provides an efficient plug-in solution for long-tailed recognition, benefiting from its lightweight background-aware feature rectification and training-only foreground-guided augmentation.

D.9 Additional Results for Eigen Spectral Density of Hessian

This section presents additional results on the eigen spectral density of the Hessian for ResNet models trained with CE, BCL, and BCL+OFBD on CIFAR-100-LT. As shown in Fig. 10, we separately analyze the Hessian spectra of head and tail classes. A smaller largest eigenvalue $\lambda_{\max}$ and trace $Tr(H)$ generally indicate a flatter loss landscape and better optimization stability.

Compared with the CE baseline, BCL tends to reduce the sharpness of the loss landscape to some extent, especially for tail classes, suggesting that contrastive learning helps improve the optimization behavior under long-tailed distributions. However, its Hessian spectrum still shows relatively large eigenvalues, indicating that the optimization landscape remains sharp for certain class groups. In contrast, BCL+OFBD consistently produces lower $\lambda_{\max}$ and $Tr(H)$ for both head and tail classes. This suggests that BCL+OFBD effectively smooths the loss landscape, rather than only benefiting tail

Table 13: Comparison with attention modules on CIFAR-100-LT ($r = 100$).

Method	Overall $\uparrow$	Many $\uparrow$	Med. $\uparrow$	Few $\uparrow$
CBAM [92]	50.43	67.80	48.90	31.90
BAM [59]	50.88	68.70	49.30	32.00
Coordinate Attention [23]	50.83	69.40	49.10	31.30
Triplet Attention [108]	50.67	67.90	50.00	31.40
BFR (Ours)	52.97	66.96	53.65	35.86

Table 15: Integration with additional long-tailed methods.

Method	Overall $\uparrow$	Many $\uparrow$	Med. $\uparrow$	Few $\uparrow$
MDCS [103]	53.15	67.09	56.09	33.47
MDCS + OFBD	54.39	68.31	55.26	37.14
SADE [100]	49.20	61.31	51.14	32.80
SADE + OFBD	51.01	57.89	53.49	40.10

Table 17: Sensitivity to the number of selected proposals K .

K	Overall Acc. (%) $\uparrow$
2	53.87 $\pm$ 0.34
3	53.93 $\pm$ 0.22
4	53.94 $\pm$ 0.20
5	53.89 $\pm$ 0.19
6	53.96 $\pm$ 0.19

Table 14: Robustness of B_u under input perturbations.

Input	BG AUROC $\uparrow$	Δ vs. Clean
Clean	0.7182	0.0000
Brightness, factor 1.2	0.7054	-0.0128
Contrast, factor 1.2	0.7051	-0.0131
Gaussian noise, std 0.05	0.7038	-0.0144
Gaussian blur, $k = 5$, $\sigma = 1.0$	0.7016	-0.0166

Table 16: Feature-layer analysis for the Region Selector.

Layer	Res.	S-FG $\uparrow$	S-BG $\downarrow$	AUROC $\uparrow$	Overall $\uparrow$	Few $\uparrow$
layer1	32 $\times$ 32	0.1618	0.1441	0.7285	53.98	41.00
layer2	16 $\times$ 16	0.1694	0.1280	0.7375	54.04	40.97
layer3	8 $\times$ 8	0.1709	0.1026	0.8644	54.12	41.15

Table 18: Masking analysis based on background-aware score B_u .

Masking	Overall Acc. (%) $\uparrow$	Δ Acc.
None	58.475	0.000
Random 20%	53.850	-4.625
High- B_u 20%	56.655	-1.820
Low- B_u 20%	45.140	-13.335

classes. The flatter Hessian spectrum further supports that BCL+OFBD mitigates background-biased optimization and leads to more stable learning under long-tailed training.

D.10 Visualization of LossLandscape

Fig. 11 visualizes the loss landscapes of head and tail classes for ResNet models trained with CE, BCL, and BCL+OFBD on CIFAR-100-LT. The first row corresponds to head classes, and the second row corresponds to tail classes. Compared with CE and BCL, BCL+OFBD tends to produce a flatter and smoother loss landscape for both class groups. Such a tendency is more pronounced for tail classes, where the loss landscape under long-tailed training is usually sharper and more irregular. The results indicate that OFBD helps stabilize the optimization process and mitigates the unfavorable optimization behavior caused by data imbalance.

D.11 Comparison with Attention Modules

We compare BFR with representative attention modules under the same BCL-based setting on CIFAR-100-LT ($r = 100$). As shown in Table 13, BFR achieves better overall and Few-shot performance, demonstrating its effectiveness in background bias.

D.12 Robustness of Background Score under Input Perturbations

We evaluate the robustness of B_u under different image perturbations on ImageNet-LT. As shown in Table 14, BG AUROC remains above 0.70 across all perturbations, indicating stable background-aware estimation under appearance variations.

D.13 Integration with Multi-Expert Long-tailed Recognition Frameworks

We integrate OFBD into representative multi-expert long-tailed recognition frameworks while keeping their original architectures and training objectives unchanged. As shown in Table 15, OFBD consistently improves different multi-expert baselines, demonstrating its compatibility and effectiveness across existing long-tailed recognition frameworks.

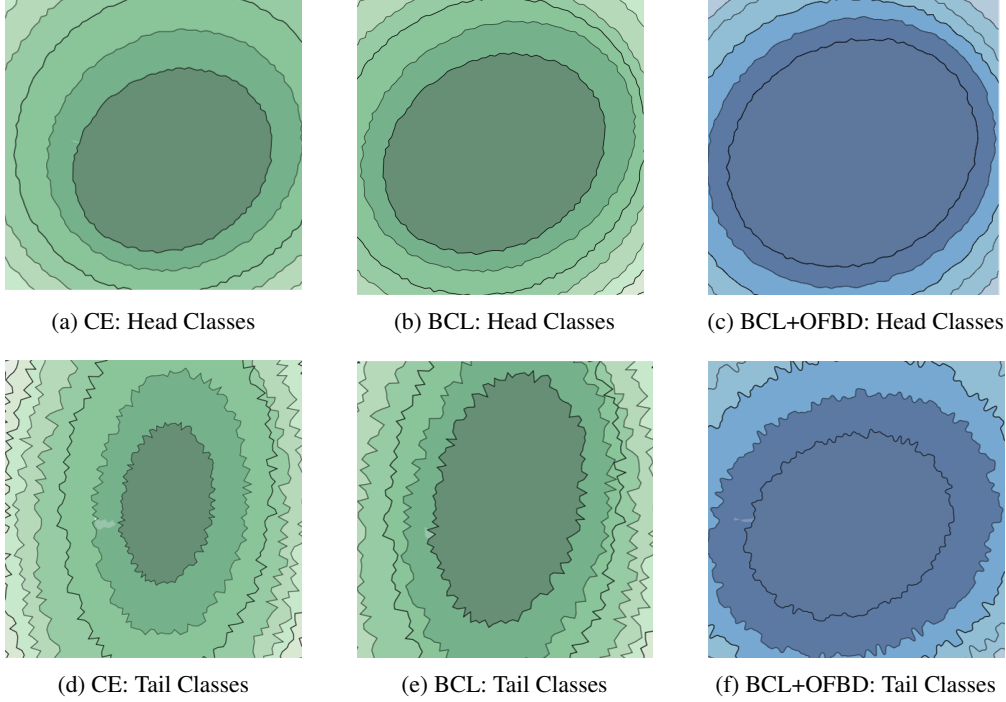


Figure 11: Visualization of loss landscape for head and tail classes of ResNet models trained with CE, BCL, and BCL+OFBD on CIFAR-100 LT respectively.

D.14 Region Selector Feature Layer Analysis

We analyze the effect of different feature layers used by the Region Selector. As shown in Table 16, deeper layers provide stronger foreground-background separation, and layer3 achieves the best discrimination capability.

D.15 Sensitivity Analysis of Region Proposal Number

We evaluate the sensitivity of K in the Region Selector. As shown in Table 17, OFBD maintains stable performance across different K values, demonstrating robustness to proposal selection.

D.16 Masking Analysis of Background-aware Score

We perform masking experiments based on B_u to analyze the semantic meaning of high-score regions. As shown in Table 18, masking high- B_u regions causes limited degradation, while masking low- B_u regions leads to significant performance drops, validating the effectiveness of BFR.

E Limitation

A key consideration of OFBD lies in its reliance on accurate foreground localization and feature separation. While our current FG-CutMix and BFR designs work effectively on standard visual benchmarks, scenarios with severe occlusion, extremely small objects, or highly entangled foreground-background cues may benefit from enhanced region selection strategies or adaptive rectification. Addressing these cases represents a natural direction for extending OFBD, highlighting opportunities for future work without undermining its demonstrated effectiveness on long-tailed datasets.